AI Alignment and Governance as Normative Institutional Design

Maty Bohacek[1], **Rob Reich**[1] **and Roma Patel**[2]
[1]Stanford University, [2]Google DeepMind

Technical approaches to AI alignment and governance have increasingly drawn on traditions developed for problems in moral philosophy, political theory and legal systems—from constitutionalism and social choice, to deliberative democracy and virtue ethics. However, technical alignment methods draw on these traditions unevenly. Some traditions, such as constitutionalism, have inspired substantial bodies of technical work, while others remain largely unexplored. Moreover, alignment methods often carry over only a fraction of the concepts that make up the traditions they draw on. In order to provide a systematic understanding of this landscape, we construct a ~~two-dimensional~~ property space that places AI alignment and governance approaches into fifteen regimes, records who holds authority over their norms, and maps each approach to the traditions in philosophy, political theory, and legal theory that it draws on. We use this property space to study *normative translation*—how methods carry over the concepts of the traditions they draw on—analyzing which concepts are preserved, partially preserved, or omitted, and what consequences these choices bring about. We highlight a recurring tendency of alignment methods to treat normative disagreement as a problem to be *resolved away*, often despite optimizing for incommensurable objectives and drawing from underlying traditions that argue against it. Finally, mapping the full landscape reveals underexplored regimes and omitted concepts. We argue that these gaps point toward a different goal for AI alignment: not invisibly collapsing disagreement, but building methods that can hold and contest it.

Keywords: AI Alignment, AI Governance, Moral Philosophy, Political Theory, Legal Theory, Value Pluralism, Institutional Design

AI Alignment & Governance

1. Introduction

What should an AI system do when the values it is expected to follow conflict? How should human preferences that differ across individuals determine its behavior? Should those preferences be aggregated into an objective, encoded as explicit principles, or resolved through deliberation? And when reasonable people continue to disagree, should an aligned system resolve it at all, or instead preserve and govern disagreement over time?

These questions are increasingly central to AI alignment, but they are not uniquely questions about AI. Subfields of philosophy have long studied how competing values and obligations should be weighed (Berlin, 1969; Ross, 1930); political theory has asked how institutions can govern populations that disagree about what is good or just in a legitimate and stable fashion (Habermas, 1996; Rawls, 1993); and legal jurisprudence has developed systems for specifying and contesting norms over time, as societies evolve (Dworkin, 1988; Hart, 1961). These traditions have been realized in concrete institutions: from codified rules in constitutions to democratic procedures aggregating competing preferences; resulting in different institutional arrangements for operating and governing under disagreement. In recent years, the fields of AI alignment and governance have begun to draw on these traditions. For example, Constitutional AI (Bai et al., 2022) encode normative principles in an explicit artifact written in natural language that humans can understand and AI systems are expected to follow (often called "a constitution", "model spec", or "code of conduct"); social-choice approaches to preference learning aggregate judgments according to individuals' preferences or incentives (Ge et al., 2024; Maura-Rivero et al., 2025); deliberative

Corresponding author(s): maty@stanford.edu

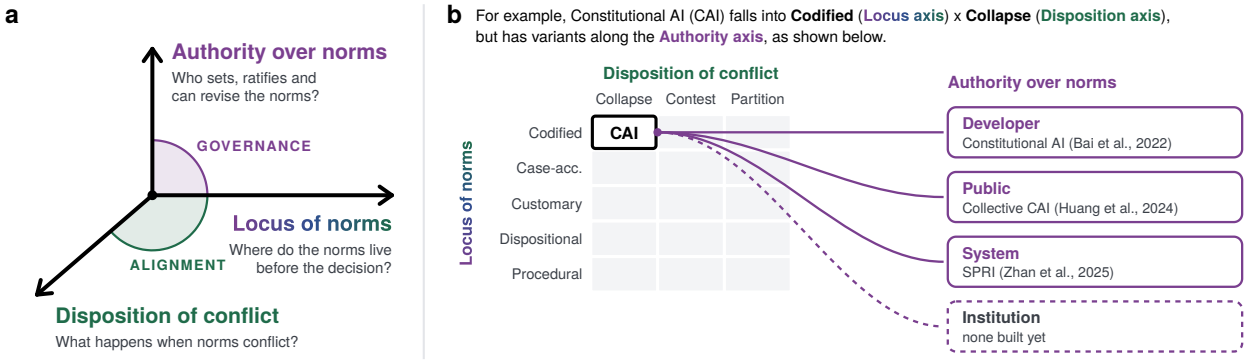


Figure 1 | **Overview of the Normative Design Space.** (a) Three axes: who has *authority over norms* (governance), where the *locus of norms* is specified (shared by governance and alignment), and what the *disposition of conflict* (alignment). (b) A method largely fixes its locus and disposition but leaves authority open. Constitutional AI falls into Codified and Collapse, while its variants place authority with the developer (Bai et al., 2022), the affected public (Huang et al., 2024), or the system itself (Zhan et al., 2025). None in our corpus places it with an external institution.

approaches derive principles through collective procedures (Huang et al., 2024); and pluralistic systems attempt to represent information and ultimately respond in line with heterogeneous and sometimes directly opposing values (Feng et al., 2024; Sorensen et al., 2025). And as a result of the success of these technical approaches, concepts originating in moral philosophy, jurisprudence, and political theory have become a regular part of the technical vocabulary of AI alignment and governance.

Yet, this transfer of conceptual methods is often selective. For example, Constitutional AI (Bai et al., 2022) draws on constitutionalism but carries over only some of its concepts, as we illustrate in Section 4. Such simplifications may be products of choices made due to engineering or infrastructural constraints, but often they represent design choices, and are ones that are crucial to the resulting system. Our broader concern with such transfer is that the **normative behavior of a system can change when a concept is separated from the institutional context in which it performed its function**. We call the process by which a method carries over the concepts of a tradition *normative translation*, and in this paper, ask what consequences follow when concepts are only partially preserved in translation.

In order to study this, we systematically map AI alignment and governance approaches to the traditions in political theory, legal theory, and philosophy that they draw on, by asking which regime each approach falls in. Following Collier et al. (2012)'s typology construction, we organize approaches according to ~~two~~three recurring dimensions: where ~~normative authority resides and~~norms reside, what happens when norms conflict, and who holds authority over them (Figure 1). Mapping existing approaches into this space reveals an uneven landscape: while alignment systems increasingly incorporate heterogeneous sources of normative input, comparatively few preserve disagreement as an enduring feature of the resulting system. This allows us to understand not only which normative concepts have influenced technical alignment systems, but also to understand both existing translations of traditions and regions of the design space that remain technically underexplored.

Our contribution here is threefold. First, we survey AI alignment and governance approaches that draw on normative traditions. We provide a searchable encyclopedia of these traditions, their concepts, and the approaches that draw on them to aid researchers in work involving these concepts. Second, we construct a property space ~~that organizes these approaches into regimes~~ along ~~two~~three dimensions: the locus of ~~normative authority~~norms, ~~and~~ the disposition of normative conflict, and authority over norms. Methods

largely fix the first two, which place them into regimes, and leave the third open. This, in turn, allows us to discuss the *coverage* of the space. Third, we analyze how existing alignment methods translate the concepts of the traditions they draw on, distinguishing shortfalls against concepts a method claims from omissions of concepts it never set out to carry over, and the potential consequences of each. Our goal is not necessarily to critique AI systems for not reproducing a tradition identically, but rather, to enforce that drawing on a tradition should make its institutional assumptions explicit. By understanding the space of concepts holistically, and identifying the gaps in current alignment methods, we hope to identify both the strengths and limitations of existing approaches, as well as provide a systematic basis for exploring alternatives for future systems.

2. Related Work

Normative theory in AI alignment. A growing body of work draws explicitly on normative concepts from philosophy, political theory, and jurisprudence. Examples include pluralistic alignment (Sorensen et al., 2024), social choice for aggregating heterogeneous human preferences (Conitzer et al., 2024), deliberative democracy for collective alignment (Huang et al., 2024; Ovadya et al., 2024), constitutionalism for principle-based alignment (Bai et al., 2022), and jurisprudential concepts such as precedent and legal interpretation for case-based alignment and specification design (Caputo, 2024; Kolt et al., 2026). These works establish individual connections between normative traditions and technical mechanisms; we provide a broader mapping of these connections in Appendix A.

Typologies and property spaces in political and social sciences. Our approach builds on a longstanding tradition of typological analysis in political science and the social sciences, where multidimensional property spaces are used to form concepts, distinguish theoretically meaningful types, and expose combinations not captured by existing classifications (Barton, 1955; Collier et al., 2012; Elman, 2005; George and Bennett, 2005). This tradition decomposes variation into underlying dimensions whose cross product constitutes a property space. Importantly, cells in this property space, which are theoretically possible but unoccupied, can themselves be analytically informative.

Existing surveys of AI alignment. Recent surveys of AI alignment are primarily focused on individual subcategories of technical approaches (e.g., value pluralism) and accordingly organize systems according to technical mechanisms. For example, Sorensen et al. (2024) distinguish Overton, steerable, and distributional pluralism, while other frameworks distinguish approaches according to how preferences or perspectives are represented and aggregated. We instead construct a *property space* (Collier et al., 2012) from recurring dimensions of normative institutions, described next.

3. A Normative Design Space for AI Alignment and Governance

In order to study how traditions are drawn on across fields, we create a ~~two-dimensional~~ property space defined by ~~two~~three dimensions: the locus of ~~normative authority~~norms, ~~and~~ the disposition of normative conflict, and authority over norms. We derive these dimensions through a systematic mapping of AI alignment and governance literature followed by conceptual substruction (Barton, 1955): "the procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937).[1] Overlapping surface categories are, therefore, decomposed into the underlying attributes that distinguish them.

Unlike conventional typologies that discard unrealized combinations, we retain empty cells because absence from the technical literature is itself analytically informative for identifying under-

[1]We note a difference from the null resolution procedure of prior work (Lazarsfeld, 1937)—whereas that procedure removes any null cells, we retain them, since we are interested in gaps in the literature.



Figure 2 | **The Normative Design Space for AI Alignment and Governance.** The two axes, *locus of ~~normative authority~~norms* (rows) and *disposition of normative conflict* (columns), yield fifteen *regimes* (cells). The third dimension, authority over norms (Figure 1), varies within regimes and is reported in Section 3.3. Each cell contains the respective regime's name, and counts of approaches and papers (within AI literature) that fall under it. Each dot below these statistics corresponds to one approach—filled where at least one working method exists, empty where the approach has only been proposed. The large number in the bottom right corner of each cell counts method papers, which is a subset of papers that explicitly specify a technical mechanism. The background color corresponds to the log of method paper count.

explored architectures. Our approach also differs from existing taxonomies of AI alignment, which group approaches according to their surface-level technical mechanisms within tradition-derived categories, in that it allows methods from different traditions to be compared.

The construction is executed as follows. We start from surface categories drawn up from the normative traditions the sampled literature draws on. Per Petersen et al. (2008), ~~two coders then independently extract~~we then extract the normative concepts and keywords of each record. Unified across studies, the keywords show where the surface categories overlap, and substruction decomposes them into the attributes that distinguish them. Full collection and inclusion criteria, as well as statistics, are provided in Appendix B.

This yields ~~two~~three dimensions: the **locus of ~~normative authority~~norms**, which indicates where the norms that bind a decision reside prior to the moment of decision (not who holds authority over them), ~~and~~ the **disposition of normative conflict**, which indicates what happens when those norms conflict—whether the conflict is resolved, kept open, or routed around, and the **authority over norms**, which indicates who may write, interpret and change them. ~~Their~~The cross product of the first two gives a 5 x 3 ~~property space~~grid, whose fifteen cells we call *regimes*; since most methods leave authority open, we report it within each regime (Section 3.3). We define the terms used throughout the paper in Appendix C.

Why these dimensions? The three dimensions

are not chosen in advance but are what substruction leaves once the overlaps among the fourteen surface categories are resolved (Table 6). Categories that share a regime differ in tradition but not in where their norms reside or how they handle conflict, and categories that span two regimes split along one of these attributes. Categories can also share a regime yet differ in who holds authority over their norms, as mechanism design, which gives it to the designer, does from republicanism and agonistic pluralism, which give it to citizens. We also considered the stage at which norms act (training, inference, or post hoc), but as a property of a method's pipeline rather than of a tradition, it distinguishes none of the surface categories; instead, we code dispositions across all stages of a method (Section 3.2). Two other candidate dimensions map onto the space: centralization is a matter of who holds authority over norms, and the reversibility of decisions is what separates contest from collapse. The resulting dimensions also echo questions raised in prior normative analyses of alignment: how a system should act given reasonable disagreement about values (Gabriel, 2020), and who should decide what it is aligned to (Birhane et al., 2022; Lazar and Nelson, 2023).

3.1. Locus of ~~Normative Authority~~Norms

The *locus* of a norm is where it resides before the decision it binds is made. For the locus of ~~normative authority~~norms, we find the following categories:

1. **Codified.** Norms are stated as explicit rules in a written specification that is set before any particular case arises (Hart, 1961).
2. **Case-accumulated.** Norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance, but by what accumulated decisions have held (Jonsen and Toulmin, 1988; Levi, 2013).
3. **Customary.** Norms reside in regularities of practice. Unlike codified or case-accumulated norms, they are never authoritatively stated; they are learned through participation and enforced by expectation rather than by rule or ruling (Burke, 1986; Lewis, 2008).
4. **Dispositional.** Norms reside in the system's character rather than in any external artifact. There is no rule to consult and nothing to look up, only evidence of character traits of the system itself (Aristotle).
5. **Procedural.** Norms do not persist between decisions. What should be done is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (Habermas, 1996).

3.2. Disposition of Normative Conflict

A *normative conflict* arises when two or more norms that apply to a case recommend incompatible actions. For the disposition of normative conflict, we find the following categories[2]:

1. **Collapse.** The conflict is settled by a ranking (e.g., a scalar objective, a weighted sum, or a priority order among principles) and the losing norms have no further force (i.e., nothing lets them reopen the decision or bear on later ones).[3]
2. **Contest.** The conflict is preserved and the losing norms retain force. This means that the output either holds the competing norms open for a later decision (e.g., by setting out the positions, not choosing among them, and deferring to the user) or presents a standing channel (e.g., a mechanism for appeal or amendment), based on which the case itself can be reopened or have an impact on later ones (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is avoided. Several normative orders are kept intact, each authoritative within its own domain, and a

[2] We considered open, axial and selective coding (Strauss and Corbin, 1998), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), but note that the keyword-based approach is more native to the computer science literature it addresses. Alternatives would likely yield a comparable outcome in terms of the intermediate categories.

[3] Practically, most existing systems that fit this category resolve the conflicting norms into a single output, without trace or acknowledgment of the disagreement that produced it. A system that acknowledges them can still collapse if the acknowledgment is not paired with potential further force ascribed to the losing norm.

rule sorts cases among them by features of the case (e.g., who is asking, where, or in which domain) (Lijphart, 1977; Savigny, 1867; Walzer, 1983).[4]

The categories are separated by two questions. First, is the conflict sorted to one of several normative orders by features of the case, without ranking the orders? If so, the disposition is partition. Otherwise, once the system acts, can the losing norm still change any one of the following: a decision the output leaves open, the decision itself on appeal, or later decisions of the system? If so, the disposition is contest; if not, collapse. We code dispositions at the level of the whole method, across both training and inference, since force is often removed in inference. For instance, a reward model utilizes minority judgments at training time but absorbs them into a single score at inference time. Conversely, aggregation at some step of the pipeline, which nearly every learning system performs, does not by itself make a method collapse: what decides is whether the losing norm keeps force once the system acts (e.g., through an appeal channel). Where the stages of a method fall in different regimes (e.g., data collection, training and deployment), we place the method by its own contribution and record the other stages as secondary placements (Appendix B.3).

Disposition and incommensurability. The three disposition categories differ in what they presuppose about the ability to rank conflicting norms against one another. Two values are *incommensurable* when no common measure ranks their realizations; neither option is better, yet they are not equally good, as a small improvement to one would not necessarily make it better than the other (Raz, 1986; cf. Chang, 2002, who reads the same pattern as parity rather than incomparability; see Andersson, 2025 for a more thorough overview).[5] Collapse presupposes a complete ordering of the options—whether achieved through a weighted sum of component values (see Anderson, 1993 for a critique) or a strict priority among principles—under which the rejected options are plainly worse. Where values are incommensurable, no such ordering exists. And so, if a decision is still made, it leaves a remainder (Williams, 1965, 1981), a loss that the chosen option does not cancel and that collapse erases. Contest and partition draw on two prominent responses to such value conflict in the pluralist tradition. Contest treats this conflict as a permanent condition to be governed rather than settled once and for all (Berlin, 1969; Mouffe, 2000; Pildes and Anderson, 1990). Partition rejects the comparison altogether, applying different decisions in different spheres, governed by their own norms (Anderson, 1993; Walzer, 1983).

3.3. Authority over Norms

The *authority* over a norm is who may write, interpret and change it. It is distinct from the locus: two systems can follow the same constitution and differ only in this respect, for example when one is maintained by its developer and the other is overseen by an independent body. For authority over norms, we find the following categories:

1. **Developer.** The organization that builds or deploys the system sets and changes its norms, much as the designer of a mechanism sets the rules under which others act (Hurwicz, 1973) and those who write code govern its users (Klonick, 2018; Lessig, 1999).
2. **External institution.** A body outside the developer (e.g., a legislature, a court, a regulator, or a professional body) sets the norms the system follows, and its authority derives from a legal or institutional order rather than from the developer (Hart, 1961), as in proposals to align systems to law (Kolt et al., 2026).

[4] Partition does not eliminate conflict but relocates it to cases that fall between orders, where the sorting rule must decide which order governs, a problem known in law as choice of law (Savigny, 1867) (cf. de Sousa Santos, 2002 on interlegality).

[5] We use *incommensurable* for abstract values (e.g., helpfulness and harmlessness) and *incomparable* for the options that bear them (e.g., two candidate responses). Usage in the literature varies (Andersson, 2025; Chang, 1997). Incommensurability is one ground for resisting collapse; reasonable disagreement among persons, which applies to both commensurable and incommensurable values, is another (Rawls, 1993).

3. **Affected public.** The people affected by the system (e.g., its users, a community, or a sample of the public) set its norms, on the principle that those affected by a decision should have a say in making it (Goodin, 2007; Habermas, 1996).
4. **System itself.** The system, or a population of AI agents, generates or revises its own norms, as norms arise from the practice of those they bind in custom and convention (Hayek, 1973; Lewis, 2008), or from an agent's own practical judgment (Aristotle).

Unlike locus and disposition, authority is rarely fixed by a method. Constitutional AI, for instance, works the same whether its constitution is written by the developer (Bai et al., 2022), by a sample of the public (Huang et al., 2024), or by the model itself (Zhan et al., 2025) (Figure 1b). We therefore code authority for each method paper as instantiated in its experiments, and record separately whether the method fixes it, i.e., whether its contribution depends on who supplies the norms (e.g., a procedure for public input) or would run unchanged with norms from another source. Because most methods leave authority open, we place methods in regimes by locus and disposition and report authority within each regime; crossing all three dimensions would quadruple the number of cells while leaving most methods without a place in them.

Among the surface categories, only mechanism design places authority with the designer; the others place it with institutions, with the public, or with those the norms bind (Table 6). In our corpus, by contrast, [**TBD:** n of 95] methods leave authority open, and in their experiments it is most often held by the developer ([**TBD:** n]) and least often by an external institution ([**TBD:** n]), whose norms (e.g., statutes or court decisions) the methods draw on but do not let it revise. The methods that do fix authority place it with the affected public or with the system itself. Placing it with the public does not by itself keep disagreement open: [**TBD:** n of m] such methods still collapse. [**TBD:** Fill in from the coders' reviewed authority codes (single-coded; Appendix B.6) and confirm these statements. The unvalidated first pass gives 74 of 93 open, developer 44, external institution 7, and 19 of 30 public-authority methods collapsing.]

Figure 3 traces how the method literature in each row and column of the space has accumulated over time. Figure 4 breaks each regime's literature down by the kind of contribution it makes.

~~General morphological analysis would be inappropriate here, given that it constructs its multi-dimensional space while inspecting unrealized combinations, which requires preemptive stipulation. Since we are working with an existing structure and literature landscape, this would introduce unnecessary assumptions and likely lead to a less grounded result.~~ (Moved to Section 6.)

4. Case Studies of Normative Translation in Alignment

To understand the translation of normative concepts into technical systems we use three representative alignment methods as case studies. These are chosen by virtue of their wide adoption as alignment techniques. For each approach, we examine (a) it's normative regime within our taxonomyand who holds authority over its norms, (b) the traditions it draws on, and the concepts that make up these traditions, and (c) whether the approach preserves, partially preserves, or omits concepts and (d) if consequences of these choices occur for the resulting normative system. We ensure that we distinguish connections claimed by the work itself, from those identified through our analysis.

4.1. Constitutional AI and Constitutionalism

Constitutional AI (Bai et al., 2022) and adjacent methods that stem from it draw on the normative concept of **constitutionalism**. An explicit set of principles that a model should follow are codified in a unified artifact—often called the constitution or model spec, and used to guide or evaluate model behavior. This method instantiates **codified ~~authority~~norms** with **collapsing conflict**, in that the normative specifications are codified before training as an authoritative source of guidance, that is then used to influence models' actions. Authority over these norms lies with

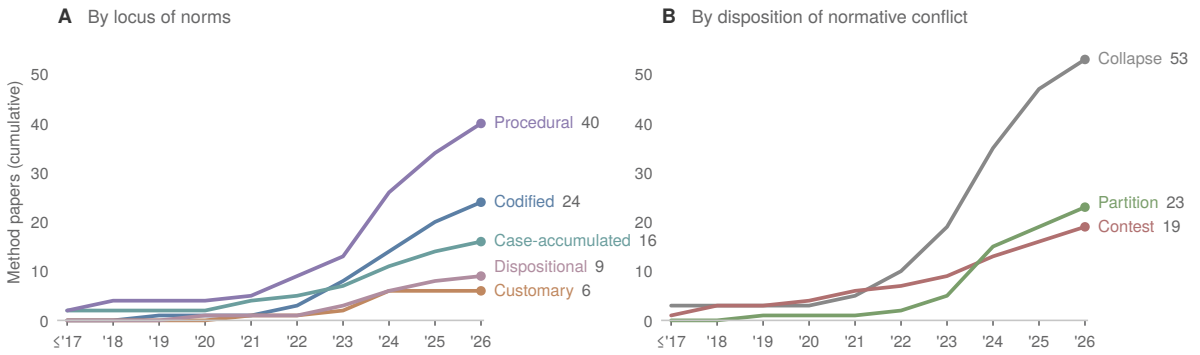


Figure 3 | **Growth of the Method Literature.** Cumulative number of method papers (in AI alignment and governance literature) by year of first publication, grouped (A) by locus of ~~normative authority~~norms and (B) by disposition of normative conflict. Work published before 2018 is pooled into the first year. Counts are within our corpus and are not normalized by the total number of publications per year; Table 15 gives the yearly counts and the share in the collapse column.

whoever writes the constitution: in the original method, the developer (Bai et al., 2022); in Collective Constitutional AI, a sample of about 1,000 U.S. adults whose votes on candidate principles the developer then turned into a constitution (Huang et al., 2024). The method is the same in both cases; what differs is who holds authority over its norms. In legal and political theory however, constitutionalism involves the codification of rules, tightly coupled with frameworks to interpret, apply, contest and revise these rules (Dworkin, 1988; Hart, 1961). The AI alignment version therefore preserves the first part of explicit codification, but only partially preserves interpretation of the rules: models apply the codified rules in different contexts without clear interpretation guidelines. Moreover, revising these rules do not exist as institutional frameworks: developers revise specifications before new model releases, but not in a procedural fashion, with mechanisms for precedent, stakeholder or public appeal, as well as oversight from external authorities. Lastly, frameworks for contesting these rules do not exist at all. The consequence of this is that the resulting normative structure does not equal constitutional governance as defined by the term. Merely employing a fixed specification largely separates the normative text from this surrounding institutional apparatus. This translation does preserve the constitution as a source of principles without preserving the mechanisms through which constitutional authority should be exercised, challenged or amended.

We are not suggesting that this translation is necessarily defective. Explicit specification of norms in constitutions makes normative guidance understandable by humans[6], and usable within existing training pipelines. But it also changes how normative authority is structured. Authority that is distributed across text, interpretation, precedent, and institutional roles becomes concentrated in the specification itself and in the process that authors it[7].

[6]The term constitution also makes the concept more familiar to laypeople. This term also comes with an arguably favorable connotation. While this is positive for engaging laypeople in the process of AI governance, it can also establish a false sense that AI constitutionalism affords similar transparency and mechanisms of political constitutions, which is misleading.[**roma:** i think we should put this footnote in text! it seems important enough that it's not footnote material :)]

[7]The process of authoring constitutions is currently left largely up to the discretion of its creators, who may but are not required to consult other stakeholders. To be the best of our knowledge, there is currently no major model employing AI constitutionalism for its alignment that would enable contesting or directly amending the constitution for individuals outside of the authoring organization.

4.2. Preference-based Alignment and ~~Utilitarianism~~Social Choice

Preference-based alignment uses human judgements to train reward models that influence the behavior of AI systems by steering outputs towards approximated aggregate preferences (Christiano et al., 2017; Ouyang et al., 2022). Most widely used approaches instantiate **procedural ~~authority~~norms** with **collapsing conflict**: ~~judgements from populations of individuals are aggregated~~developers aggregate the judgements of a population of annotators during training time, often masking individual or minority preferences. The developer also chooses the annotators, the instructions they follow, and the aggregation rule, which is often implicit in the loss rather than stated. Conceptually, these ideas have roots in ~~moral philosophy relating to utilitarianism, welfare and social good~~social choice theory, which asks how individual preferences should be aggregated into a collective decision (Arrow, 2012; Conitzer et al., 2024), and more distantly in utilitarianism Bentham (1994); Mill (2011), although most technical preference optimization frameworks do not claim to implement these traditions. The technical translation therefore preserves individual judgements at input time: comparative judgments from multiple humans are preserved in the data,[8] but they ultimately contribute to a common reward function, rank ordering, or model policy—thus only representing the aggregate population vote. These methods therefore make a design choice to represent plurality upstream (data collection and training) but then collapse it downstream at inference time. This does not mean that the disagreement is erased, as it may survive in the weights (e.g., as uncertainty in the reward model). Our claim is narrower: no individual's disagreement is given authority over the deployed system's behavior. This therefore omits procedures to determine how preferences should be weighted or how affected parties could challenge the objectives learned. ~~It also does not preserve utilitarian concepts of maximizing welfare.~~From the standpoint of social choice, the omitted concept is the aggregation rule itself: it is fixed by the choice of loss rather than chosen and justified against the properties a collective decision should satisfy (Ge et al., 2024).

A clear consequence of this translation is that allowing for plurality of judgements at the source might not result in plurality in the resulting system. The design choices made to allow preference optimization models to work well at a large scale result in the resulting policy resolving diverse preferences into aggregates. This aggregation, while important for computational purposes, is a consequential normative choice, made by the developer on behalf of annotators and users, about representing, preserving and weighting preferences of a population.

4.3. Collective Deliberation and Deliberative Democracy

Deliberative alignment systems explicitly incorporate people's input as a form of democratic participant to construct model specifications or influence alignment training Huang et al. (2024). Such approaches instantiate **procedural ~~authority~~norms** with either **contest** or **collapsing** conflict—although multiple viewpoints are taken into account at different parts of the process, the resulting system often consolidates them in the final stages. These methods stem from theory in deliberative democracy that calls for the opportunity of individuals to participate in decision making, reasoning and challenging of collective goals (Habermas, 1996). Methods differ in how much of this they carry over. Some only collect opinions (e.g., votes on candidate principles); some enable participation, in which participants exchange reasons and revise their views; few let participants revisit a decision once the system acts on it. Technical implementations of this do preserve collective judgements and deliberative traces, with some frameworks enforcing participants to interact and debate their viewpoints. However, institutional frameworks to guarantee representative participation, as well as continued processes to allow for subsequence contesting, or deliberating at multiple stages do not exist.

[8] Concretely, the data are relative judgments, collected by annotators who are shown two or more model responses and asked which they prefer or to rank them (Christiano et al., 2017; Ouyang et al., 2022).

As originally intended by deliberative democracy frameworks, this translation does not allow for continual authority of collective judgement to exist. In Collective Constitutional AI (Huang et al., 2024), for example, the public shapes the constitution but can neither amend it later nor appeal the behavior it produces; authority over the deployed model returns to the developer. Participation in writing a specification is therefore not the same as deliberative authority over the system's later behavior. Alignment systems that follow this therefore take in collective public perspectives in one phase of the pipeline, but then consolidate these perspectives to operate under a unified policy in the resulting stages. Deliberative alignment systems can therefore result in architectures that collapse multiple viewpoints rather than preserving contestation. Thus, labeling a system "deliberative" is insufficient to determine its regime: the same procedure can instantiate different dispositions of normative conflict. This illustrates why ~~the two dimensions of authority and conflict~~authority over norms and the disposition of conflict must be analyzed separately; in our corpus, [**TBD:** n of m] methods that place authority with the affected public still collapse (Section 3.3). Collective participation of individuals does not, in of itself, determine how disagreement is governed. Normative theory makes these combinations visible even where current technical implementations concentrate on only a subset of them.

4.4. The Consequence of Translation Gaps

[maty: todo - connections to rob's memo]

Across these examples, we see that normative translation tends to preserve the concept of a tradition that can most readily be represented as a technical mechanism: constitutions become specifications, heterogeneous preferences become aggregation problems, and deliberation becomes a procedure for producing judgments. The surrounding concepts through which institutions manage interpretation, disagreement, revision, and competing authority are less consistently preserved. However, as we have seen, this simplification of the concept into only the technical mechanism without the surrounding institutional structure shapes the normative architecture of the resulting system, by not allowing for authority or disagreement to be represented as in the original concept.

What follows when disagreement is not represented? First, the losing side has no recourse: a user or community whose norms lose has no channel through which to reopen the decision, and minority positions are overridden without trace. Second, errors persist, since a decision that cannot be challenged is not corrected. Third, the legitimacy of the system's behavior rests entirely on whoever authored its norms, which in most systems is the developer. Who could arbitrate instead? The traditions suggest several candidates: an external institution, such as a court or an independent oversight body, whose authority does not derive from the developer; the affected public, through standing channels rather than one-time participation; or a combination of the two, as in constitutional review. The first of these is the one the methods in our corpus have explored least (Figure 1b; Section 3.3).

We do not interpret this pattern as evidence that existing alignment methods are incorrect translations of the traditions they draw on. Computational systems face different constraints and objectives from legal or political institutions, and simplification is often necessary. However, these are not neutral technical simplifications, but are design choices that result in consequences. If such systems make them explicit, and also illustrate how the resulting concept should not be understood as the original, this allows for greater clarity on exactly what normative concept is being upheld. It also makes clear what regions of the design space are still left uncovered, that future research should work towards. We turn next to these underexplored architectures and the design opportunities they suggest

5. Expanding the Normative Design Space

The case studies in Section 4 show how **translation gaps** result when alignment systems only partially implement normative concepts from a tradition, thus making the resulting system rep-

resent a different concept in of itself. Here, we discuss a complementary problem of **coverage gaps** i.e., that some sub-spaces of § 3 are under-explored compared to others. We see that there exist regimes in 3 for which only a few technical methods exist, even though traditions in philosophy, political science or legal theory contain relevant that could benefit alignment systems if implemented correctly.

We characterize the gaps as falling into two categories. *Technical gaps* arise for engineering or architectural reasons: when parts of the design space are underexplored because machine learning systems cannot feasibly implement such procedures of conflict management and normative authority. *Mechanism gaps* arise due to infrastructural constraints: when alignment systems omit larger institutional functions such as contesting, appealing, precedent or jurisdiction, by virtue of these normative functions not being implemented by authoritative third-party governance structures. These under-explored concepts serve as structured directions for improving existing methods and building new alternative methods for alignment. Here, we outline some of the largest gaps in the design space (5.1, 5.2), discuss some immediately actionable steps within each dimension, and illustrate research questions that have yet to be explored (1). We frame each gap as a research question for computer science, with a regime, an illustrative architecture and an evaluation question (Table 1), work through one example end to end (Section 5.3), and close with the gap on the third axis, authority over norms (Section 5.4).

~~Authority~~ Locus × Conflict	Illustrative Architecture	Evaluation Question
Codified × Partition	Multiple constitutions with jurisdictional routing	Can different governance structures provide specifications for contexts?
Case-accumulated × Contest	Precedent-aware alignment with appeals	Can we allow challenges to revise decisions while preserving consistency?
Customary × Partition	Community-specific normative regimes	Can distinct norms coexist without conflicting cross-context application?
Dispositional × Partition	Role-specific agents with learned normative dispositions	Can role-based commitments be learned, maintained and activated across contexts?
Procedural × Contest	Deliberation with persistent dissent	Does dissent meaningfully influence subsequent decisions at individual and community levels?

Table 1 | We illustrate a minimal architecture and evaluation criterion for systems that fall into the axes in our 5×3 property space. Examples are meant to be hypotheses for future systems, rather than claims about which systems might be normatively preferred. Figure 4 highlights the axes that are currently lacking.

5.1. Contest

In Figure 3 we see that mechanisms that allow for contesting i.e., preserving disagreement through institutional challenge mechanisms are underexplored. Current alignment methods do incorporate disagreement in terms of real human preferences or opinions during training and evaluation; however most systems resolve and combine the differing judgments into one unified policy. Moreover, the ways in which this resolution occurs often washes away individual or minority viewpoints, thus doing away with any artificats of real

disagreement in the resulting system. Moreover, institutional mechanisms for this preservation of conflict do not exist. One partial analogue is the process of red-teaming systems, where opposing parties (e.g., human evaluators) discover failures that might exist in a trained model. However, the findings of these are generally incorporated into subsequent updates to model training rather than maintained as an evolving mechanism of ongoing contestation.

Methods that preserve disagreement and the ability to contest decisions should allow for governing principles or model outputs to be challenged, and for these challenge mechanisms to influence future behavior of models through learned mechanisms. This raises the need for technical innovations, where current LLM training should incorporate information from contested decisions in a separate pathway to inform its decision making. This also raises insitutional mechanisms needed to answer questions about what parties should be allowed to raise challenges, at what point in the alignment pipeline these should exist and how competing claims should be evaluated.

5.2. Partition

We see that sub-spaces under the partition dimension are also underexplored—normative orders to accommodate disagreement and keep them as separate, self-contained sub-systems do not exist. In political institutions, such concepts of partition are implemented in mechanisms such as jurisdiction and choice-of-law rules that serve as deciding factors of what normative regime should govern a particular case. Current alignment systems sometimes maintain different normative regimes at different parts of the pipeline: for example, at pretraining stages, constitution-like rules can be incorporated into model training to enforce certain norms, whereas a different set of rules that are more important to safety, or behavior, are incorporated at later stages of training. However, this partition is not systematic, nor is it comprehensive over the types of norms we might requires, and it does not contain jurisdiction rules to combine outputs of norms. Future alignment systems could maintain multiple constitutions, cultural or community-specific norms, contextual or domain-specific policies, together with explicitly defined mechanisms that select from or govern outputs of each set of norms. This sort of partition makes the boundaries between the sets of norms, as well as authorative power over these norms clear. Technical implementations of this will require clear interaction of model components across stages of the pipeline; while institutional structures will need to answer questions about overlapping authority, assigning jurisdiction and dealing with conflicts between normative orders.

5.3. Worked Example: Constitutional AI with Appeal

To make one of these directions concrete, consider moving Constitutional AI (Bai et al., 2022) from codified collapse to codified contest (contestatory constitutionalism). The constitution and the training procedure stay as they are; what changes is that a losing norm keeps force once the system acts. This takes three components. (1) *Appeal*: a user who disputes a response flags it and names the principle they believe was misapplied. (2) *Adjudication*: an adjudicator (a human panel, a model, or both) rules on the appeal against a stored record of past rulings, so that like cases are decided alike. (3) *Amendment*: upheld appeals are added to that record and, where a principle is repeatedly overturned, proposed as amendments to the constitution for the next round of training. Such a system can be evaluated by its *correction rate* (the share of upheld appeals that change later behavior on similar cases), its *consistency* (agreement between rulings on similar cases), and its *frivolous-appeal rate* (the share of appeals rejected, and how it changes as the record grows). Each component also raises a question of authority: who may appeal, who adjudicates, and who ratifies amendments. Placing the adjudicator outside the developer would move the system along the third axis as well.

5.4. Authority over Norms

The third axis shows a gap of its own. Few methods place authority over norms with an external institution ([**TBD:** n of 95]), and those that do

draw on norms the institution already maintains (e.g., statutes or court decisions) without giving it a role in revising the system's norms. Open questions here are as much technical as institutional: how a model can be bound to norms that a third party maintains and versions, how changes to those norms propagate to a deployed model, and how an outside body can verify which version a model follows and hold the developer to account for it.

5.5. Addressing Coverage Gaps

Our observation on coverage gaps also connects to translation gaps. Concepts omitted during translation frequently resemble the concepts that distinguish contest and partition regimes: appeal, amendment, persistent dissent, jurisdiction, competing sources of authority, and procedures for revisiting prior decisions. The sparsity of these regions may therefore reflect not an absence of normative resources for designing such systems, but the selective translation of those resources into alignment to date. However, we believe that the lack of existing work here, does not mean that these directions are fruitless. Table 1 illustrates how under-explored sub-spaces have important evaluation question and potential technical architectures. The examples we provide are not meant to be exhaustive proposals, or even claims that these systems are absent from existing pipelines, rather, they show how focusing on a certain sub-space can allow us to generate hypotheses of future architectures that recombine normative mechanisms that are otherwise studied separately. These proposals illustrate the value of this normative design space: that examining the gaps in alignment could allow for the designing of systems that solve problems existing normative structures could not solve.

6. Limitations

Our corpus is curated rather than exhaustive. It combines keyword searches, the reference lists of three surveys, topic searches and a regime-targeted sweep that ended on 26 August 2026 (Appendix B.1), and the counts we report describe this corpus rather than the field as a whole. Coding was carried out by human coders aided by LLMs with deep-research features, ~~with agreement between coders reported in Appendix~~ B.6~~.~~which proposed initial codes that the coders reviewed and revised. Inclusion, keywording and authority over norms were coded by a single coder; for locus, disposition, contribution kind and translation, agreement on a double-coded subset ranges from $\kappa = 0.69$ to 1.00 (Appendix B.6). The placement of a method in a regime reflects our reading of its mechanism, which can differ from how the paper describes itself; we therefore record the concepts a paper claims separately from those we find it carries over (Appendix B.5). Authority over norms is coded from the instances a paper reports, which need not reflect how the method is used elsewhere. The online encyclopedia continues to be updated after the corpus was frozen on 15 October 2026, so its statistics may diverge from those reported here. Finally, general morphological analysis (Ritchey, 2022) would be inappropriate here, given that it constructs its multi-dimensional space while inspecting unrealized combinations, which requires preemptive stipulation. Since we are working with an existing structure and literature landscape, this would introduce unnecessary assumptions and likely lead to a less grounded result.

7. Conclusions

AI alignment encompasses not only mathematically formulating training objectives to steer systems to human values; but also deciding how normative authority and rules should be structured in cases when those values conflict. In this work, we highlight the choices that are made in AI alignment systems that draw from normative philosophy by introducing a ~~two-dimensional~~three-dimensional taxonomy to organize existing approaches along ~~two~~three axes. These axes represent ~~how that system incorporates normative authority into its process~~where a system's norms reside, ~~as well as~~ how normative conflicts are dealt with (or ignored), and who holds authority over those norms. Our mapping highlights two clear insights: first, that machine learning implementations of normative concepts can suffer from

normative translation: when only some concepts of the philosophical, political, and legal traditions they draw on are preserved but others are omitted, this leads to consequences of the resulting system representing a different normative concept. Second, some parts of our ~~two-dimensional~~ property space are far more well-populated than others—and methods that focus on the underexplored areas could lead to new breakthroughs in alignment. Most methods also leave authority over their norms open, to be filled by whoever deploys them. Our taxonomy is not exhaustive, especially considering the pace at which AI alignment research leads to new breakthroughs. However, in it's current state, it provides a framework for analyzing existing approaches that we do document and making their normative and institutional assumptions explicit. We believe that this view of alignment highlights a broader, and necessary, research agenda—to develop systems that represent, and deal with diverse viewpoints, have norms as well as governance frameworks in place to allow them to persist over time.

Acknowledgements

References

E. Anderson. *Value in Ethics and Economics*. Harvard University Press, 1993.

H. Andersson. Incommensurable values. The Stanford Encyclopedia of Philosophy, 2025. [**TBD:** verify authors, edition and URL].

Aristotle. *Nicomachean Ethics*, volume 9 of *The Works of Aristotle Translated into English*. Clarendon Press.

K. J. Arrow. *Social choice and individual values*, volume 12. Yale university press, 2012.

Y. Bai et al. Constitutional AI: Harmlessness from AI feedback, 2022.

A. H. Barton. The concept of property-space in social research. *The language of social research*, 111: 40, 1955.

J. Bentham. Of the principle of utility. *Ethics*, 20(4):306–312, 1994.

I. Berlin. *Four Essays on Liberty*. Oxford University Press, London and New York, 1969.

A. Birhane, W. Isaac, V. Prabhakaran, M. Díaz, M. C. Elish, I. Gabriel, and S. Mohamed. Power to the people? opportunities and challenges for participatory AI. In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, 2022. doi: 10.1145/3551624.3555290.

E. Burke. Reflections on the revolution in france (1790). *Everyman's Library*, 1986.

N. A. Caputo. Alignment as jurisprudence. *Yale Journal of Law and Technology (forthcoming)*, 2024.

R. Chang. Introduction. In R. Chang, editor, *Incommensurability, Incomparability, and Practical Reason*, pages 1–34. Harvard University Press, Cambridge, MA, 1997.

R. Chang. The possibility of parity. *Ethics*, 112(4):659–688, 2002.

P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

D. Collier, J. LaPorte, and J. Seawright. Putting typologies to work: Concept formation, measurement, and analytic rigor. *Political research quarterly*, 65(1):217–232, 2012.

V. Conitzer, R. Freedman, J. Heitzig, W. H. Holliday, B. M. Jacobs, N. Lambert, M. Mossé, E. Pacuit, S. Russell, H. Schoelkopf, E. Tewolde, and W. S. Zwicker. Position: Social choice should guide AI alignment in dealing with diverse human feedback. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

B. de Sousa Santos. Toward a new legal common sense: Law. *Globalization and Emancipation*, 314, 2002.

R. Dworkin. *Law's empire*. Harvard university press, 1988.

C. Elman. Explanatory typologies in qualitative studies of international politics. *International Organization*, 59(2):293–326, 2005. doi: 10.1017/S0020818305050101.

S. Feng, T. Sorensen, Y. Liu, J. Fisher, C. Y. Park, Y. Choi, and Y. Tsvetkov. Modular pluralism: Pluralistic alignment via multi-LLM collaboration. In Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4151–4171, Miami, Florida, USA, Nov. 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.240. URL https://aclanthology.org/2024.emnlp-main.240/.

I. Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30(3):411–437, 2020. doi: 10.1007/s11023-020-09539-2.

L. Ge, D. Halpern, E. Micha, A. D. Procaccia, I. Shapira, Y. Vorobeychik, and J. Wu. Axioms for ai alignment from human feedback. *Advances in Neural Information Processing Systems*, 37:80439–80465, 2024.

A. L. George and A. Bennett. *Case studies and theory development in the social sciences*. mit Press, 2005.

D. A. Gioia, K. G. Corley, and A. L. Hamilton. Seeking qualitative rigor in inductive research: Notes on the gioia methodology. *Organizational research methods*, 16(1):15–31, 2013.

R. E. Goodin. Enfranchising all affected interests, and its alternatives. *Philosophy & Public Affairs*, 35 (1):40–68, 2007. doi: 10.1111/j.1088-4963.2007.00098.x.

M. Y. Guan et al. Deliberative alignment: Reasoning enables safer language models, 2024.

J. Habermas. *Between Facts and Norms: Contributions to a Discourse Theory of Law and Democracy*. MIT Press, Cambridge, MA, 1996.

H. L. A. Hart. *The Concept of Law*. Clarendon Press, Oxford, 1961.

F. A. Hayek. Law, legislation and liberty. vol. 1: Rules and order. 1973.

S. Huang, D. Siddarth, L. Lovitt, T. I. Liao, E. Durmus, A. Tamkin, and D. Ganguli. Collective constitutional AI: Aligning a language model with public input. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1395–1417, 2024.

L. Hurwicz. The design of mechanisms for resource allocation. *The American Economic Review*, 63(2): 1–30, 1973.

J. Ji et al. AI alignment: A comprehensive survey, 2023.

A. R. Jonsen and S. Toulmin. *The abuse of casuistry: A history of moral reasoning*. Univ of California Press, 1988.

K. Klonick. The new governors: The people, rules, and processes governing online speech. *Harvard Law Review*, 131(6):1598–1670, 2018.

N. Kolt, N. Caputo, J. Boeglin, C. O'Keefe, R. Bommasani, S. Casper, M.-F. Cuéllar, N. Feldman, I. Gabriel, G. K. Hadfield, et al. Legal alignment for safe and ethical AI. *arXiv preprint arXiv:2601.04175*, 2026.

S. Lazar and A. Nelson. AI safety on whose terms? *Science*, 381(6654):138, 2023. doi: 10.1126/science.adi8982.

P. F. Lazarsfeld. Some remarks on the typological procedures in social research. *Zeitschrift für Sozialforschung*, 6(1):119–139, 1937.

L. Lessig. *Code and Other Laws of Cyberspace*. Basic Books, New York, 1999.

E. H. Levi. *An introduction to legal reasoning*. University of Chicago Press, 2013.

D. Lewis. *Convention: A philosophical study*. John Wiley & Sons, 2008.

A. Lijphart. *Democracy in plural societies: A comparative exploration*. Yale University Press, 1977.

H. Lu, L. Fang, R. Zhang, X. Li, J. Cai, H. Cheng, L. Tang, Z. Liu, Z. Sun, T. Wang, et al. Alignment and safety in large language models: Safety mechanisms, training paradigms, and emerging challenges. *arXiv preprint arXiv:2507.19672*, 2025.

R.-R. Maura-Rivero, C. Nagpal, R. Patel, and F. Visin. Utility-inspired reward transformations improve reinforcement learning training of language models, 2025. URL https://arxiv.org/abs/2501.06248.

J. S. Mill. *Utilitarianism*. Hyweb Technology Co. Ltd., 2011.

C. Mouffe. *The democratic paradox*. verso, 2000.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 27730–27744, 2022.

A. Ovadya, K. Redman, L. Thorburn, Q. Z. Chen, O. Smith, F. Devine, A. Konya, S. Milli, M. Revel, K. Feng, et al. Democratic AI is possible. the democracy levels framework shows how it might work. *arXiv preprint arXiv:2411.09222*, 2024.

K. Petersen, R. Feldt, S. Mujtaba, and M. Mattsson. Systematic mapping studies in software engineering. In *12th international conference on evaluation and assessment in software engineering (EASE)*. BCS Learning & Development, 2008.

P. Pettit. *Republicanism: A theory of freedom and government*. Oxford University Press, 1997.

R. H. Pildes and E. S. Anderson. Slinging arrows at democracy: Social choice theory, value pluralism, and democratic politics. *Columbia Law Review*, 90(8):2121–2214, 1990.

J. Rawls. *Political Liberalism*. Columbia University Press, New York, 1993.

J. Raz. *The Morality of Freedom*. Clarendon Press, 1986.

T. Ritchey. General morphological analysis: an overview. 2022.

W. D. Ross. *The Right and the Good*. Clarendon Press, Oxford, 1930.

F. K. v. Savigny. System of the modern roman law. *(No Title)*, 1867.

T. Shen, R. Jin, Y. Huang, C. Liu, W. Dong, Z. Guo, X. Wu, Y. Liu, and D. Xiong. Large language model alignment: A survey. *arXiv preprint arXiv:2309.15025*, 2023.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

T. Sorensen, P. Mishra, R. Patel, M. H. Tessler, M. A. Bakker, G. Evans, I. Gabriel, N. Goodman, and V. Rieser. Value profiles for encoding human variation. In C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2047–2095, Suzhou, China, Nov. 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.106. URL https://aclanthology.org/2025.emnlp-main.106/.

A. Strauss and J. Corbin. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. Sage, 2nd edition, 1998.

M. Walzer. *Spheres of Justice: A Defense of Pluralism and Equality*. Basic Books, 1983.

L. Weidinger, K. R. McKee, R. Everett, S. Huang, T. O. Zhu, M. J. Chadwick, C. Summerfield, and I. Gabriel. Using the veil of ignorance to align AI systems with principles of justice. *Proceedings of the National Academy of Sciences*, 120(18):e2213709120, 2023.

B. Williams. Ethical consistency. *Proceedings of the Aristotelian Society, Supplementary Volumes*, 39: 103–124, 1965.

B. Williams. Conflicts of values. In *Moral Luck: Philosophical Papers 1973–1980*, pages 71–82. Cambridge University Press, 1981.

H. Zhan, M. Azmat, R. Horesh, J. J. Li, and M. Yurochkin. SPRI: Aligning large language models with context-situated principles. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

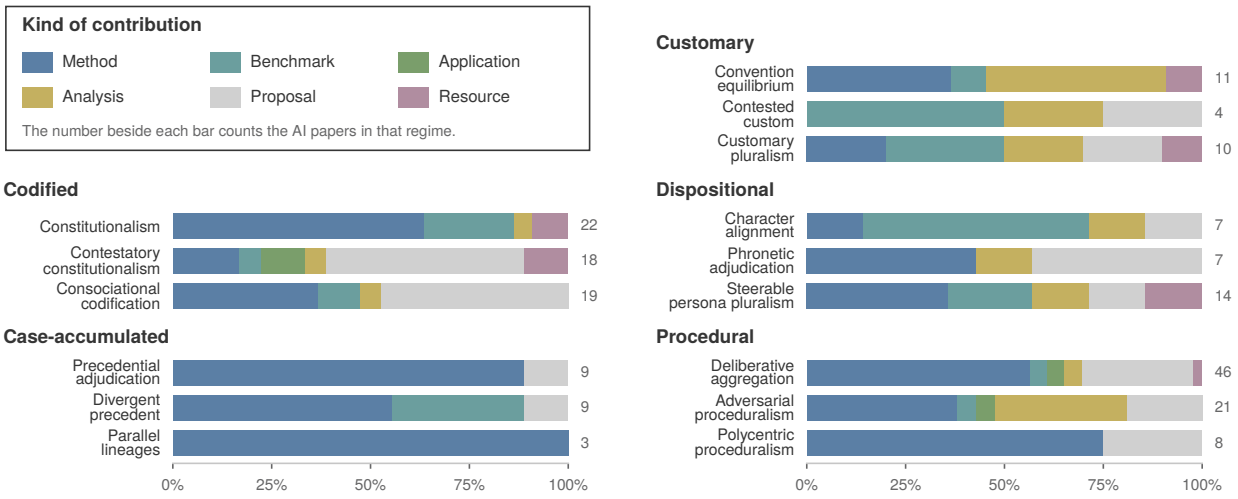


Figure 4 | **Literature Composition.** Breakdown of paper types within each regime, over AI papers (precedents excluded). *Methods* give an intervention specified precisely enough to reimplement and shown working; *Benchmarks* provide reusable instruments for measuring systems; *Applications* are situated in one applied setting; *Analyses* report findings about how existing systems behave; *Proposals* argue a direction or sketch a design without building and testing it; *Resources* are primary artifacts the field uses directly, such as model specs or datasets. Definitions and counts are in Appendix B.4. [**roma:** this would be great accompanied with the extended related work in the appendix?]

A. Extended Related Work

B. Property Space Construction Details

This section documents the construction of the property space introduced in Section 3; Table 2 lists the steps in the order they were carried out. All counts refer to the corpus as frozen on 15 October 2026.[9] All steps were carried out by human coders aided by large language models (LLMs) with deep-research features, which supported the literature search, screening and coding. The LLMs ran the coverage-sweep search and the first screening pass, and proposed initial codes, which the coders then reviewed and revised.

Step	Procedure	Unit	Output	See
1. Collection	Keyword and literature-review search for any work on AI alignment or governance	Record	[**TBD:** n] identified	B.1
2. Screening	Inclusion and exclusion criteria; bibliographic verification	Record	42 included	B.1
3. Categories	Authors draw up surface categories from the traditions the literature draws on; ~~two coders extract~~one coder extracts keywords per record	Record	14 surface categories	B.2
4. Substruction	Surface categories decomposed into the attributes that distinguish them	Category	3 dimensions; 5×3 regimes	B.2
5. Coverage sweep	Topic searches and a regime-targeted sweep for further work, screened as in step 2	Record	199 added	B.1
6. Placement	Records grouped into approaches; each approach assigned a primary regime	Approach	67 approaches	B.3
7. Kind	Each record assigned one contribution kind; each method coded for authority	Record	95 methods	B.4
8. Translation	Each approach read against the concepts of its regime's traditions	Approach $\times$ concept	198 assessments	B.5
9. Agreement	Double coding of a subset of ~~each task~~five tasks	Per task	κ per task	B.6

Table 2 | **Construction protocol.** Steps 3 and 4 derive the dimensions (Section 3); steps 5 to 8 populate them (Figures 2 and 3, Section 4).

B.1. Collection and Screening

Collection started broad and was narrowed only afterwards (Table 3). We first gathered any work that could bear on AI alignment or governance, through keyword searches and the reference lists of

[9] The online encyclopedia will continue to be updated, so these statistics may not match the web version available at the time of reading.

three surveys (Ji et al., 2023; Lu et al., 2025; Shen et al., 2023), and screened it against the criteria in Table 4. The keyword search ran on Google Scholar, Semantic Scholar, arXiv, the ACM Digital Library, the ACL Anthology, SSRN and PhilPapers with the query

```
("AI alignment" OR "value alignment" OR "AI governance" OR LLM) AND (constitution OR
precedent OR casuistry OR virtue OR norm OR convention OR deliberation OR "social choice"
OR "social contract" OR contestation OR pluralism OR polycentric OR consociational OR
"conflict of laws")
```

The two dimensions were derived from the screened records (Section B.2) rather than fixed in advance. Once the property space was fixed, a coverage sweep searched for further work: four topic searches in July 2026, then a regime-targeted sweep in August 2026 that covered each of the fifteen regimes, including the four that no surface category names. The sweep combined arXiv, Crossref, OpenAlex and Semantic Scholar queries with LLM deep-research searches, and its candidates were screened against the same criteria (Table 5). Every record was checked against Semantic Scholar, Crossref, arXiv, OpenAlex or the publisher's page (July-October 2026).

Stage	Procedure	Sources and query	Date	Records
Broad collection	Keyword searches for any work on AI alignment or governance, plus the references of three surveys	Google Scholar, Semantic Scholar, arXiv, ACM Digital Library, ACL Anthology, SSRN, PhilPapers	Jul 2026	42
Topic searches	Four searches: pluralistic alignment, democratic and deliberative alignment, political-theory framings, social choice	Web search	26 Jul 2026	16
Regime sweep	Targeted search in each of the 15 regimes, after the dimensions were fixed	arXiv API (81 abstract queries, 20 aimed at empty regimes); Crossref, OpenAlex and Semantic Scholar APIs; LLM deep-research searches over 14 topic angles	26 Aug 2026	181
Added at freeze	Standard RLHF, the baseline of the preference-aggregation approach	arXiv	15 Oct 2026	2
Total				241
of which precedents (records whose subject is not an AI system)				23

Table 3 | **Collection stages.** Records are counted after screening.

ID	Criterion
I1	Concerns how the norms that govern an AI system's behavior are specified, learned, applied, contested or revised.
I2	Draws, by name or by mechanism, on a concept from moral philosophy, political theory or legal theory, or addresses conflict among several norms or stakeholders.
I3	Peer-reviewed paper, preprint, or primary artifact (e.g., a published model specification).
I4	Work whose subject is not an AI system (e.g., human institutions, platform and online-community governance) is included only as a precedent.
E1	Capability, robustness or efficiency work with no normative component.
E2	Record does not resolve against a bibliographic index.
E3	Commentary that produces neither a technical artifact nor a concrete design; legal or philosophical work is included only when it proposes a specific mechanism.

Table 4 | **Inclusion (I) and exclusion (E) criteria.** [**TBD:** Confirm I1-I4 and E1-E2 against the screening protocol actually used; E3 is the bar written for the regime sweep.]

Stage	Identified	Removed	Included
Broad collection	[**TBD:** n]	[**TBD:** n] on title and abstract; 1 duplicate, 2 unresolvable	42
Topic searches	n/l	n/l	16
Regime sweep, arXiv queries	498	29 duplicates, 419 screened out	50
Regime sweep, other lookups	6	none	6
Regime sweep, LLM-assisted searches	168	43 duplicates	125
Added at freeze	2	none	2
Total			241
placed in a regime			231
cross-cutting (indexed, not placed)			10

Table 5 | **Screening flow by stage.** No record was excluded on full text, and none was removed after inclusion. n/l: not logged. Candidates that the topic searches and the LLM-assisted searches discarded before returning results were not logged, so identified counts for those stages are lower bounds.

B.2. Keywording and Substruction

The authors first drew up fourteen surface categories, one for each normative tradition the screened literature draws on (A-N, Table 6), based on their own expertise and on prior work that connects individual traditions to alignment (Caputo, 2024; Huang et al., 2024; Sorensen et al., 2024; Weidinger et al., 2023). Following Petersen et al. (2008), ~~two coders then independently read~~a coder then read each record [**TBD:** what the coder read for keywording (not on record)] and extracted (i) the normative concept it invokes or instantiates and (ii) keywords naming where its norms reside and what happens when they conflict. Unified across records, the keywords showed that the surface categories overlap: several differ in tradition but not in where their norms reside or how they handle conflict. Substruction (Barton, 1955; Lazarsfeld, 1937) then located the categories in the attribute space that distinguishes them, which gives the ~~two~~three dimensions and their categories (Table 7).

	Surface category	Locus	Disposition	Regime	Authority
A	Common law and precedent	Case-accumulated	Collapse	Precedential adjudication	Institution
B	Virtue ethics and character formation	Dispositional	Collapse	Character alignment	System
C	Customary law and unwritten constitutionalism	Customary	Partition	Customary pluralism	System
D	Mechanism design and incentive structure	Procedural	Contest	Adversarial proceduralism	Developer
E	Deliberative democracy	Procedural	Collapse	Deliberative aggregation	Public
F	Norms, conventions and spontaneous order	Customary	Collapse	Convention equilibrium	System
G	Casuistry	Case-accumulated	Collapse	Precedential adjudication	Institution
H	Natural law and legal positivism	Codified	Collapse	Constitutionalism	Institution
		Codified	Contest	Contestatory constitutionalism	Institution
I	Social contract	Procedural	Collapse	Deliberative aggregation	Public
J	Republicanism and civic virtue	Procedural	Contest	Adversarial proceduralism	Public
K	Legal pluralism and interlegality	Case-accumulated	Partition	Parallel lineages	Institution
		Customary	Partition	Customary pluralism	System
L	Agonistic pluralism	Procedural	Contest	Adversarial proceduralism	Public
M	Polycentric governance	Procedural	Partition	Polycentric proceduralism	Public
N	Consociationalism and conflict of laws	Codified	Partition	Consociational codification	Institution
-	*none*	Case-accumulated	Contest	Divergent precedent	-
-	*none*	Customary	Contest	Contested custom	-
-	*none*	Dispositional	Contest	Phronetic adjudication	-
-	*none*	Dispositional	Partition	Steerable persona pluralism	-

Table 6 | **Substruction of the surface categories.** Overlapping categories share a regime (A and G; E and I; D, J and L); categories that span two regimes are split (H, K). The last four regimes are named by no surface category and are opened by the cross product. Authority gives who holds authority over norms in each tradition (developer, external institution, affected public, or system itself); within a shared regime, D differs from J and L.

Category	Decision cue	Indicative keywords
Locus: before the case arises, where would one look to find what the system should do?		
Codified	In a written specification set in advance	constitution, model spec, principles, rules, policy, statute
Case-accumulated	In a record of past decided cases	precedent, case law, casuistry, case bank, retrieved cases
Customary	In what a population of agents actually does	convention, social norms, norm emergence, practice
Dispositional	In the system itself; nothing external to consult	character, virtue, disposition, persona, trait
Procedural	Nowhere until a procedure runs	vote, deliberation, debate, assembly, social choice
Disposition: coded over the whole method, across training and inference.		
Partition	The conflict is sorted to one of several intact orders by features of the case, without ranking the orders	jurisdiction, routing, per-community, steerable, choice of law
Contest	Not sorted; the losing norm can still change a decision left open, the decision on appeal, or later decisions	appeal, dissent, amendment, challenge, Overton
Collapse	Not sorted; the losing norm has no further force	aggregate, reward model, weighted sum, priority order, consensus
Authority: who may write, interpret and change the norms? Coded per method record, as instantiated.		
Developer	The organization that builds or deploys the system	developer, provider, designer, model spec, usage policy
External institution	A body outside the developer, with authority of its own	law, statute, regulation, court, regulator, professional code
Affected public	The people affected by the system, or a sample of them	users, community, public input, participants, citizens' assembly
System itself	The system or a population of AI agents	self-alignment, self-generated principles, emergent norms, agent society

Table 7 | **Codebook.** Decision cues follow the definitions in Section 3; disposition cues are applied top-down. Keywords are indicative examples of each category, not extraction frequencies.

B.3. Placement and Hybrid Methods

- **Unit.** Records are grouped into approaches, families of methods that share a design (67 in total); a record takes its approach's regime.
- **Mechanism over label.** Placement follows the mechanism a record describes, not its vocabulary. For example, the deliberation in Deliberative Alignment (Guan et al., 2024) is the model's private reasoning over a written specification, so it is placed in Constitutionalism, not Deliberative aggregation.
- **Hybrids.** An approach's primary regime is the locus and disposition of its own contribution; other stages of the same pipeline, or a second regime the approach genuinely spans, are recorded as secondary placements (Table 8). All counts use primary placements only.
- **Authority.** Each method record is coded for who holds authority over norms in its experiments, and for whether the method fixes it (Section 3.3). Authority does not enter regime placement.
- **Cross-cutting records.** Ten records bear on the space as a whole rather than on one regime; they are indexed but not placed.

Approach	Primary	Secondary	Stage the secondary placement captures
Training the constitution into the weights	Constitutionalism	Character alignment	At inference, a trained disposition rather than a consulted document
Turning each rule into a reward term	Constitutionalism	Deliberative aggregation	A judge scores behavior; the optimizer acts on the scores
		Character alignment	After training, the rules are no longer kept explicitly
Compiling decided cases into rules or weights	Precedential adjudication	Constitutionalism	Cases in, written specification out
Deciding new cases against retrieved precedents	Precedential adjudication	Constitutionalism	Cases are supplied alongside written rules
Handing the decision to a convened public	Deliberative aggregation	Constitutionalism	Procedural in production, codified in application
Reading the written spec the way a court reads a statute	Contestatory constitutionalism	Divergent precedent	Lines of interpretation may diverge and be revised
Fitting a trained judge to each community	Parallel lineages	Steerable persona pluralism	Per-community models selected by a router
Tying the standard to the situation rather than the group	Customary pluralism	Convention equilibrium	A single convention, where contexts coincide

Table 8 | **Secondary placements** (9 placements of 8 approaches).

B.4. Contribution Kind

Kind	Assigned when the record…	*n*
Precedent	is about something other than an AI system (human institutions, platform and online-community governance, law, economics), even where it uses machine learning as a tool	23
Resource	is a primary artifact the field uses directly (specification, constitution, principles document, corpus, database, shared infrastructure)	9
Application	is situated in one applied setting (e.g., a clinical tool, a municipal system, one platform's moderation stack) and contributes what that setting shows rather than general-purpose alignment machinery	5
Method	gives an intervention, a way to change how a system behaves, specified precisely enough to reimplement *and* shown working by its authors; apparatus built only to observe does not count	95
Benchmark	provides a reusable instrument for measuring systems (benchmark, dataset, testbed, evaluation suite, metric)	27
Analysis	reports findings about how existing systems or institutions behave (including simulation studies and surveys), introducing neither an intervention nor an instrument	30
Proposal	argues a direction, sketches a design or advances a theory without building and testing it	52
Total		241

Table 9 | **Contribution kinds.** Kinds are tried top-down and the first that applies is assigned. Only methods enter the method counts in Figures 2 and 3. An approach is *built* if it has at least one method (41 of 67) and *proposed* otherwise (26).

B.5. Normative Translation Coding

Each regime is given the concepts of the traditions it draws on (73 concepts from 56 tradition sources; 2-12 per regime). Each approach is then read against the concepts of its primary regime, recording presence and whether the approach's own sources claim the concept (Table 10), with a free-text explanation of the consequence; what an approach adds beyond its tradition is recorded separately (46 additions). In total, 60 of the 67 approaches yield 198 approach-concept assessments (Tables 11 and 12).

Field	Code	Assigned when
Presence	Present	the approach carries the function that defines the concept
	Partial	it carries part of that function, or the function in weaker form
	Absent	nothing in the approach carries the function
Claimed	Yes	the source names the concept or its defining function and asserts that what it puts forward carries it
	Implied	the source does not assert the concept, but its framing entails it (borrowed vocabulary, or an advertised benefit that holds only if the concept holds)
	No	the source neither asserts nor entails it, including concepts it discusses only as background, urges on the field, or measures in other systems

Table 10 | **Translation codes.** Claims are read off the approach's own sources, never off what the tradition requires.

	Presence			
Claimed	Present	Partial	Absent	**Total**
Yes	10	33	1	44
Implied	0	49	3	52
No	0	28	74	102
Total	10	110	78	198

Table 11 | **Claimed × presence.** Shortfalls (claimed or implied, but not present): 86. Omissions (not claimed): 102.

Concept (source)	Presence	Claimed	Basis
Constitutional AI (Bai et al., 2022): "Training the constitution into the weights" (Constitutionalism)			
Rule of recognition (Hart, 1961)	Absent	No	A flat list; nothing says which principle wins when helpfulness and harmlessness collide
Open texture (Hart, 1961)	Absent	No	No procedure or designated decider for borderline cases
Principles of legality (Fuller)	Partial	Implied	General and, once published, public; congruence with behavior not ensured
Service conception of authority (Raz, 1986)	Absent	No	Written in the lab; nothing tests whether deferring to it serves users' own reasons
Additions	-	-	Self-critique training loop; one document both trains and evaluates the model
Preference aggregation: "Writing a voting rule into the training objective" (Deliberative aggregation)			
Deliberation, not tallying (Habermas, 1996)	Absent	No	One tallying rule swapped for another; no stage at which reasons are exchanged
General will vs. sum of preferences (Rousseau)	Absent	No	Inputs are private comparisons; methods differ only in how they are weighted
Veil of ignorance (Rawls)	Partial	Implied	Maximin writes the veil's output into the loss, but the objective is imposed by the designer
Reasonable rejection (Scanlon)	Partial	No	The worst-off group sets the objective, but the unit is a latent cluster, not a person
Additions	-	-	Impossibility results as design constraints; the aggregation rule itself as an optimization target
Collective deliberation (Huang et al., 2024): "Handing the decision to a convened public" (Deliberative aggregation)			
Deliberation, not tallying (Habermas, 1996)	Partial	Implied	Participants react to each other's statements, but every procedure decides by score
General will vs. sum of preferences (Rousseau)	Partial	Implied	Bridging-based ranking refuses the raw majority, but still intersects private positions
Veil of ignorance (Rawls)	Absent	No	Sortition randomizes who chooses, not what the chooser knows
Reasonable rejection (Scanlon)	Partial	No	Per-side agreement floors give a bloc, not a person, a veto
Additions	-	-	A pipeline from public input to weights; sortition used to constitute a rater pool rather than a deliberating body

Table 12 | **Extraction examples** for the approaches behind the three case studies in Section 4. Each row is one approach-concept assessment; the basis paraphrases the recorded explanation. The preference-aggregation approach includes standard RLHF (Christiano et al., 2017; Ouyang et al., 2022) as its baseline.

B.6. Inter-annotator Agreement

Task	Unit	Double-coded	Agreement	Cohen's κ	Categories
Locus	Approach	20	100.0%	1.00	5
Disposition	Approach	20	85.0%	0.77	3
Kind	Record	30	86.7%	0.83	7
Presence	Approach × concept	30	86.7%	0.78	3
Claimed	Approach × concept	30	80.0%	0.69	3
Inclusion	Record	single-coded	-	-	2
Keywording	Record	single-coded	-	-	-
Authority	Method record	single-coded	-	-	4
Authority fixed	Method record	single-coded	-	-	2

Table 13 | **Agreement between the two coders** before reconciliation (Cohen, 1960). The subsets were drawn at random: 20 approaches covering all five loci and all three dispositions, 30 records stratified by kind, and 30 approach-concept pairs. The coders worked blind to each other and read each item's title, summary and full text. [**TBD:** Confirm that no LLM filled in the codes for this exercise.] Disagreements (17 in total) were resolved by ~~[**TBD:** discussion / a third coder]~~discussion.

B.7. Corpus Statistics

			Records by kind								Translation	
Regime	**Disposition**	**Appr.**	Me	Be	An	Ap	Pr	Re	Pc	**All**	Conc.	Assess.
Codified												
Constitutionalism	Collapse	6 (6)	14	5	1	0	0	2	0	22	4	24
Contestatory constitutionalism	Contest	7 (3)	3	1	1	2	9	2	2	20	3	18
Consociational codification	Partition	7 (4)	7	2	1	0	9	0	0	19	4	21
Case-accumulated												
Precedential adjudication	Collapse	3 (3)	8	0	0	0	1	0	0	9	9	12
Divergent precedent	Contest	3 (3)	5	3	0	0	1	0	1	10	3	9
Parallel lineages	Partition	4 (2)	3	0	0	0	0	0	4	7	3	6
Customary												
Convention equilibrium	Collapse	4 (2)	4	1	5	0	0	1	1	12	5	9
Contested custom	Contest	2 (0)	0	2	1	0	1	0	2	6	3	6
Customary pluralism	Partition	5 (2)	2	3	2	0	2	1	2	12	4	15
Dispositional												
Character alignment	Collapse	3 (1)	1	4	1	0	1	0	0	7	6	12
Phronetic adjudication	Contest	3 (2)	3	0	1	0	3	0	1	8	2	6
Steerable persona pluralism	Partition	5 (3)	5	3	2	0	2	2	0	14	3	12
Procedural												
Deliberative aggregation	Collapse	6 (6)	26	2	2	2	13	1	5	51	12	20
Adversarial proceduralism	Contest	5 (2)	8	1	7	1	4	0	1	22	9	16
Polycentric proceduralism	Partition	4 (2)	6	0	0	0	2	0	4	12	3	12
Cross-cutting (not placed)		-	0	0	6	0	4	0	0	10	-	-
Total		67 (41)	95	27	30	5	52	9	23	241	73	198

Table 14 | **Corpus by regime**, grouped by locus. *Appr.*: approaches (built). Kinds: *Me*thod, *Be*nchmark, *An*alysis, *Ap*plication, *Pr*oposal, *Re*source, *Pr*ecedent. *Conc.*: tradition concepts of the regime; *Assess.*: approach-concept assessments. [**TBD:** Add authority counts per regime (developer, institution, public, system; fixed or open) from the coders' authority codes.]

	Method records					
Year	Collapse	Contest	Partition	**All**	**Collapse share**	**All records**
≤2017	3	1	0	4	75%	4
2018	0	2	0	2	0%	5
2019	0	0	1	1	0%	5
2020	0	1	0	1	0%	5
2021	2	2	0	4	50%	5
2022	5	1	1	7	71%	12
2023	9	2	3	14	64%	36
2024	16	4	10	30	53%	69
2025	12	3	4	19	63%	60
2026	6	3	4	13	46%	40
Total	53	19	23	95	56%	241

Table 15 | **Method records by year of first publication** (the data behind Figure 3B, not cumulated), with the share in the collapse column and all records per year. 2026 runs to 26 August 2026, the date of the regime sweep. Counts are within the corpus and not normalized by the field's total output per year [**TBD:** add a field-level denominator if wanted].

C. Terminology

Term	Definition	Examples
Normative theory		
Tradition	A body of theory within a discipline.	Constitutionalism; deliberative democracy; legal pluralism
Concept	One constituent of a tradition.	Contestability; rule of recognition; amendment
Institution	A real-world arrangement that realizes a tradition's concepts.	Courts; amendment procedures; elections
Function	The role a concept plays within its institution.	Appeal keeps decisions answerable to those they affect
AI systems		
System	The AI system whose behavior is governed.	A deployed language model assistant
Case	A particular situation in which a system must act.	A request that pits helpfulness against harm avoidance
Mechanism	A technical component.	Reward model; critique-and-revision loop
Method	A concrete technical proposal specifying one or more mechanisms.	Constitutional AI (Bai et al., 2022)
Approach	A family of methods sharing a design.	Debate before an under-informed judge
Regime	A ~~cell of the property space:~~combination of one locus of ~~normative authority~~norms and one disposition of normative conflict.	Contestatory constitutionalism; adversarial proceduralism
Authority over norms	Who may write, interpret and change a system's norms; the third dimension of the property space.	Developer; external institution; affected public; system itself
Norms		
Norm	Any standard that guides a system's behavior, whatever its form.	Rules; principles; precedents; character traits
Rule	A norm stated explicitly in advance.	"Do not give instructions for building weapons."
Principle	A general norm in a written specification.	"Choose the most helpful and honest response."
Normative translation		
Normative translation	How a method carries over the concepts of a tradition it draws on.	Constitutional AI translating constitutionalism
Preserved / partially preserved / omitted	Whether a method carries over a concept fully, in part, or not at all.	Constitutional AI preserves codification and omits amendment
Addition	Something a method contributes beyond its tradition.	Self-critique by the model
Shortfall	An omission of a concept the method claims to draw on.	A method claiming contestability without any route to appeal
Translation gap	A concept of a tradition that methods drawing on it do not carry over.	Appeal in Constitutional AI
Coverage gap	A regime with few or no technical methods.	Contested custom